Solution Requirements

Date	03 October 2026
Team ID	TEAM-ID-TO-BE-FILLED
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users register with username, email and password (min 8 chars, stored as bcrypt hash) and log in to receive a 30-minute JWT in an HttpOnly cookie; logout blacklists the token	High
2	Authorization levels	Single role (registered user). Every planner, history and session endpoint requires a valid token; users can read only their own history	High
3	External interfaces	Google Gemini API (text + image) for recommendations; search-URL deep links to Amazon, Flipkart, IKEA, Myntra, Ajio, Swiggy, Zomato, BookMyShow, OYO, MakeMyTrip, NoBroker, BlueStone, Tanishq, CaratLane and others	High
4	Transactions processing	Each planner request is validated, sent to Gemini, checked against the budget and saved to the recommendations table; no payments are processed	High
5	Reporting	Itemised budget summary (total, planned spend, remaining), per-category calculation table, dashboard recent activity, history with full detail view, print / save	Medium
6	Business rules	Planned spend must never exceed the budget (excess items are removed); amounts are in INR; the party plan includes a contingency; image upload limited to JPG/PNG/WEBP up to 5 MB	High
7	Compliance to laws or regulations	Passwords hashed; API key kept in .env, never committed; only the data needed for the plan is stored; users can read only their own records (aligned with India's DPDP Act principles of purpose limitation)	Medium
8	Other	Offline fallback plan if Gemini is unavailable; automatic model fallback chain; input edge-case validation	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Application overhead is small; AI latency dominates	Avg app response < 2 s, max < 5 s (excl. Gemini); measured 0.10 s avg at 25 users
2	Scalability	Stateless request handling, threadpool for blocking AI calls, DB behind a thin module that can be swapped for PostgreSQL	25 concurrent users with 0 % errors (tested)
3	Security & Data Privacy	bcrypt hashing, signed JWT, HttpOnly SameSite cookie, upload verification, AI output rendered as text (no HTML injection), prompt-injection fencing	No plaintext secrets; 100 % of protected routes return 401 without a token

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
4	Reliability & Availability	Never show an empty result: fallback plan when AI fails or returns invalid / over-budget output	100 % of valid requests return a within-budget plan (43 automated tests)
5	Usability & Accessibility	Responsive layout, labelled form fields, visible focus, clear error messages, print stylesheet	Usable at 360 px width; keyboard navigable
6	Other	Maintainability and portability: modular code, .env configuration, Dockerfile	Runs with `pip install -r requirements.txt` and `uvicorn main:app`